

Edge Intelligence LAB

Lab Task 2 - Image recognition using Edge impulse

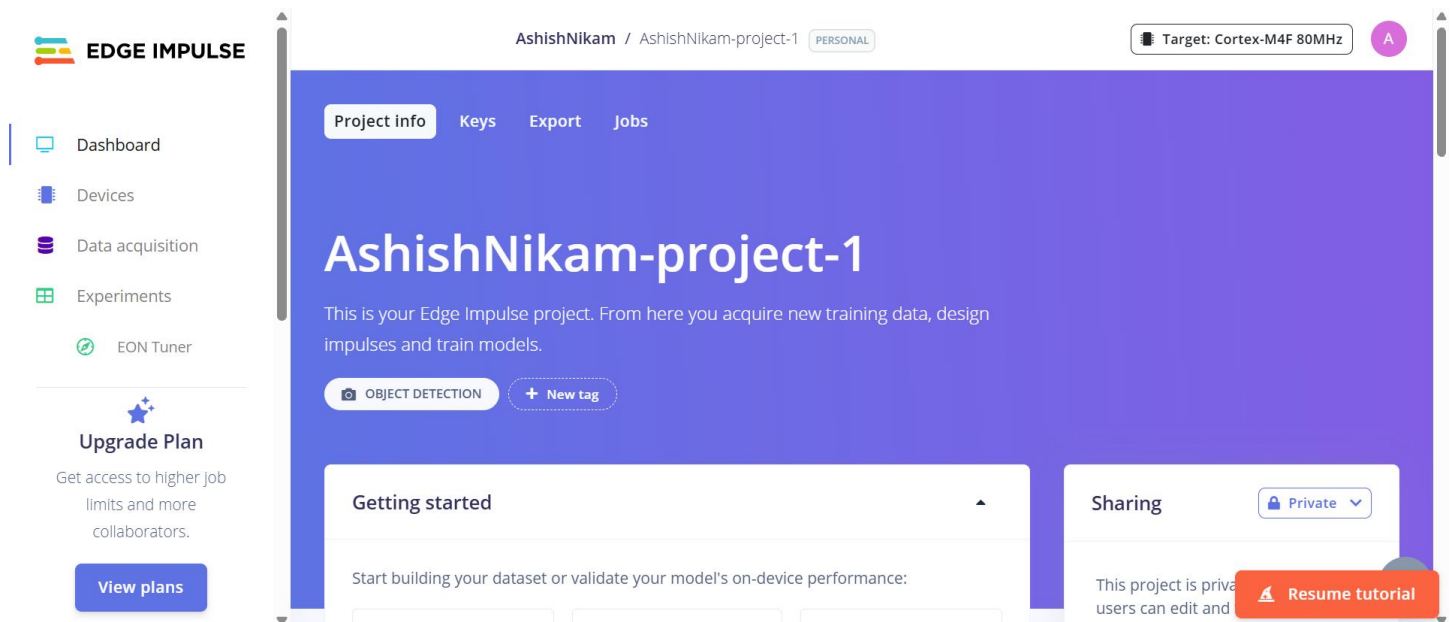
Ashish Vikas Nikam

25MML0015

GitHub Repository link :

https://github.com/ashish222001/25MML0015_Ashish.Nikam_MACSE604_EdgeLab01

1. Create account in edge impulse :



2. Go to data acquisition

EDGE IMPULSE

Dashboard
Devices
Data acquisition
Experiments
EON Tuner

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DATA COLLECTED
11 items

TRAIN / TEST SPL...
-

Collect data
[Connect a device](#) to start building your dataset.

Click on a sample to load...

Dataset

Training (7) Test (4) Post-processing (0)

SAMPLE NAME	LABELS	ADDED
Laptop bag.6ebb345f	-	Yesterday,...
Bottle.6ebb0uld	-	Yesterday,...
Keyboard.6ebav2hv	-	Yesterday,...
Mouse.6ebau3oh	-	Yesterday,...
Monitor.6ebat2o3	-	Yesterday,...

[Resume tutorial](#)

3. Choose connect data option and Scan QR using phone

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Collect new data

Collect data directly from your phone, computer, device, or development board.

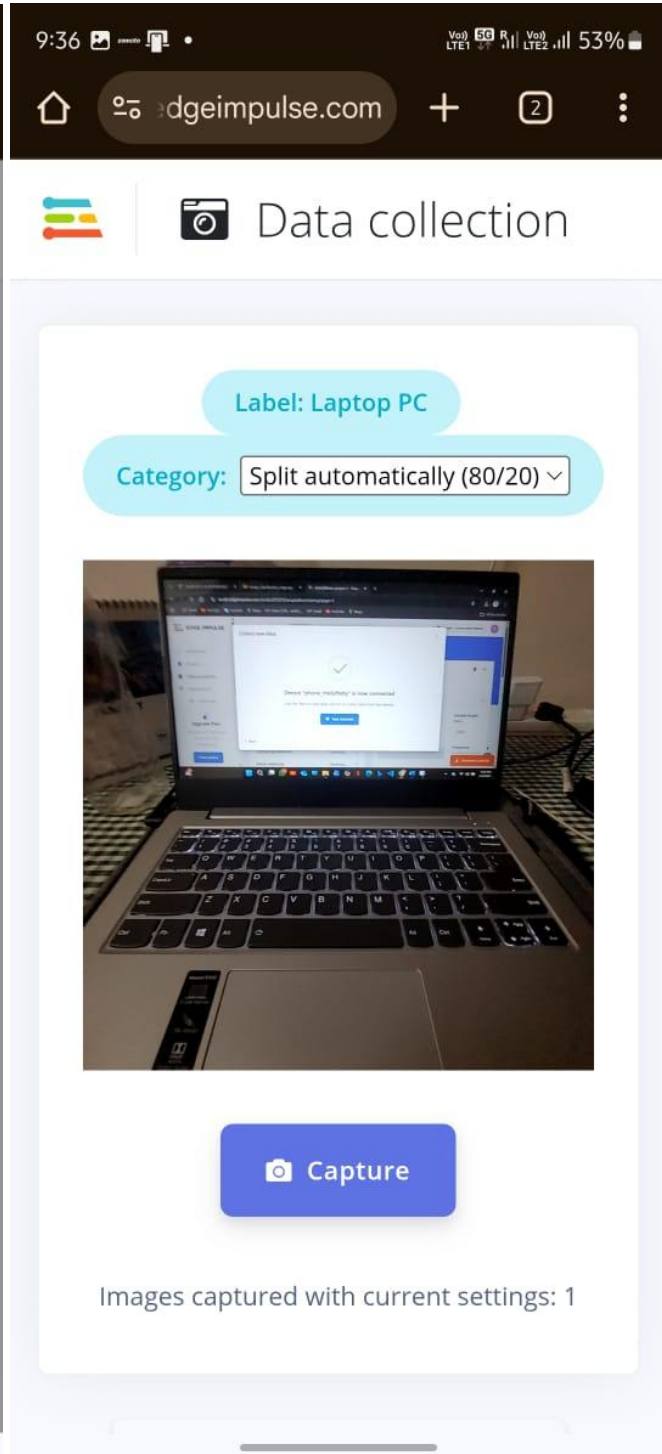
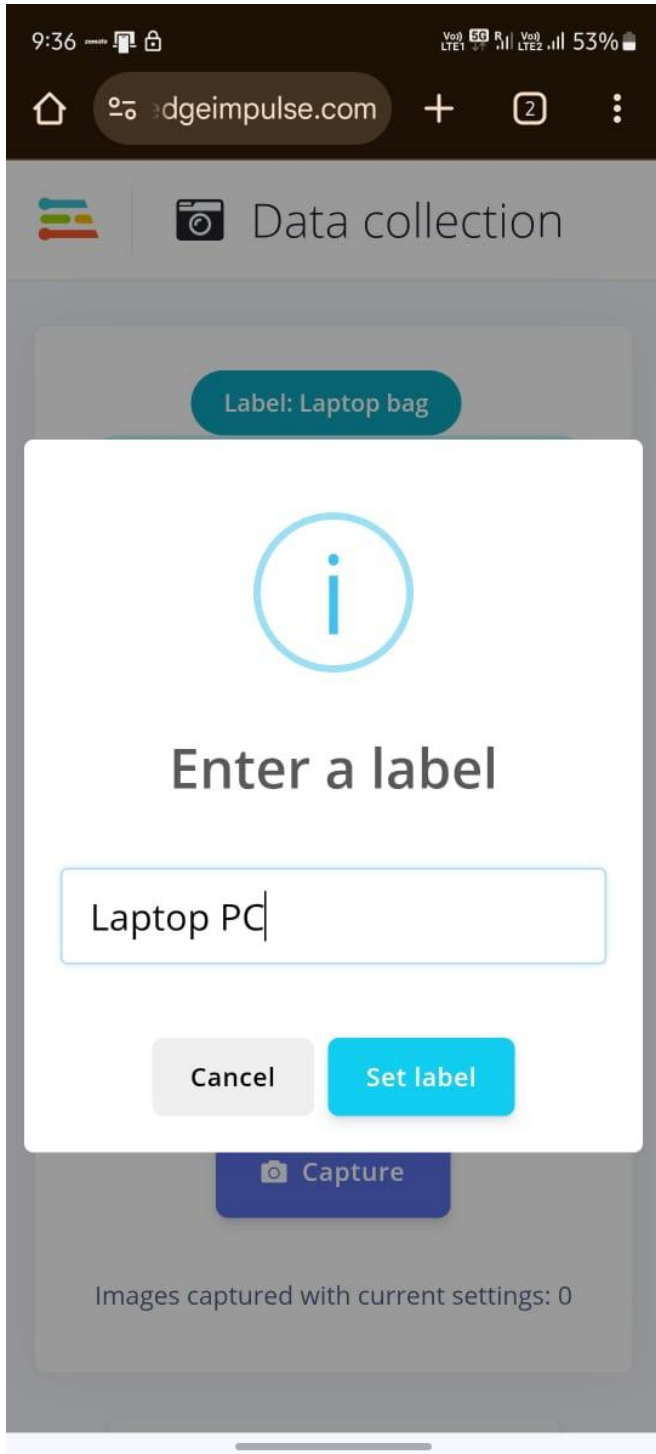
Scan QR code to connect to your phone

Connect to your computer

Connect your device or development board

Target: Cortex-M4F 80MHz

Resume tutorial

4. Select Label data before clicking a photo.

5. Split the clicked photos into train and test data

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DATA COLLECTED
12 items

TRAIN / TEST SPL...
-

Collect data
[Connect a device](#) to start building your dataset.

Click on a sample to load...

Dataset

Training (8) Test (4) Post-processing (0)

SAMPLE NAME	LABELS	ADDED
Laptop PC.6eecp676	-	Today, 21:...
Laptop bag.6ebb345f	-	Yesterday,...
Bottle.6ebb0uld	-	Yesterday,...
Keyboard.6ebav2hv	-	Yesterday,...
Mouse.6ebau3oh	-	Yesterday,...
Monitor.6ebat2o3	-	Yesterday,...

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6. Create Impulse :

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Dashboard
Devices
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An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Image data

Input axes
image

Image w... 96 Image h... 96

Resize mode
Fi

Image

Name
Image

Input axes (1)
Image
image

Object Detection (Images)

Name
Object detection

Input features
☒ Image

Output features
2 (Keyboard, Mouse)

Output features
2 (Keyboard, Mouse)

[Save Impulse](#)

[Resume tutorial](#)

7. Image : Generate Features

EDGE IMPULSE

- Dashboard
- Devices
- Data acquisition
- Experiments
- EON Tuner

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Parameters **Generate features**

Training set

Data in training set: 15 items

Classes: 2 (Keyboard, Mouse)

[Generate features](#)

Feature generation output: (0)

Feature explorer

Keyboard (blue dots), Mouse (orange dots)

[Resume tutorial](#)

8. Object Detection : model save and run

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- Dashboard
- Devices
- Data acquisition
- Experiments
- EON Tuner

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Target: Cortex-M4F 80MHz

Neural Network settings

Training settings

Number of training cycles: 60

Use learned optimizer: ☐

Learning rate: 0.001

Training processor: CPU

Data augmentation: ☒

Advanced training settings

Training output

Model version: Quantized (int8)

Last training performance (validation set)

F1 SCORE: 50.0%

Confusion matrix (validation set)

	BACKGROUND	KEYBOARD	MOUSE
BACKGROUND	100%	0%	0%
KEYBOARD	66.7%	33.3%	0%
MOUSE	-	-	-
F1 SCORE	1.00	0.50	-

[Resume tutorial](#)

Input layer (27,648 features)


FOMO (Faster Objects, More Objects) MobileNetV2 0.35

Choose a different model

Output layer (2 classes)

Save & train

Metrics (validation set)

METRIC	VALUE
Precision (non-background) ?	1.00
Recall (non-background) ?	0.33
F1 Score (non-background) ?	0.50

On-device performance ?

Engine: ? **EON™ Compiler (RAM optimized)**

 INFERE...
788
ms.

 PEAK RA...
137.7K

 FLASH U...
81.4K

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9. Model Testing : Classify

Object detection

Retrain model

Live classification

Model testing

Post-processing

Deployment

Versioning

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Target: Cortex-M4F 80MHz

This lists all test data. You can manage this data through [Data acquisition](#).

Test data Classifying...

Set the 'expected outcome' for each sample to the desired outcome to automatically score the impulse.

SAMPLE ...	EXPECTED O...	F1 SCORE	RESULT
Keyboar...	Keyboard		
Mouse.6...	Mouse		
Mouse.6...	Mouse		
Keyboar...	Keyboard		

Model testing output Cancel (0)

Merging generated features and cache...
Still merging generated features and cache... (100%)
Merging generated features and cache OK (4 windows)

Merging samples for feature explorer...
Still merging samples for feature explorer... (100%)
Merging samples for feature explorer OK
Generating features for Image OK

Classifying data for Object detection...
Classifying data for float32 model...

Resume tutorial

10. Results :

Results

Model version: ?

Unoptimized (float32)



ACCURACY ?

66.67%

Metrics for Object detection

METRIC	VALUE
Precision (non-background) ?	0.00
Recall (non-background) ?	0.00
F1 Score (non-background) ?	0.00

Feature explorer ?

- object_detection - correct
- object_detection - incorrect

